



The Hiding Hand controversy as an analytical approach in the study of urban megaevents

Antònia Casellas^{a,*}, Markku Lehtonen^b

^a Geography Department, Universitat Autònoma de Barcelona, Spain

^b Universitat Pompeu Fabra, Barcelona, Spain

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ABSTRACT

Despite growing public opposition, megaevents such as the Summer Olympics continue to proliferate both in the Global North and South, thus reshaping the built environment of cities and the living conditions of their inhabitants. This paper presents a novel analytical tool that expands the perspectives from which to explore urban megaevents. Building on the controversy among megaproject scholars around Albert Hirschman's Hiding Hand principle, we study the diverse roles that optimism can play in the justification, planning, implementation, and evaluation of megaevents. This approach has the potential to unveil key urban policy agendas underpinning the hosting of megaevents. It does so by focusing on two key aspects: first, the various types and degrees of power exercised by the involved stakeholders in their attempts to justify and support or oppose an event during the planning phase, and second, the coping mechanisms deployed by the promoters as they face unexpected constraints during the implementation phase. In doing so, the approach can help to develop more nuanced and context-specific criteria for evaluating the success of megaevents. We exemplify the approach by using the 1992 Barcelona Olympics as an illustrative case study.

1. Introduction

Because of the substantial investments in infrastructure, logistics, and organisation they require, summer Olympics are among the most transcendent types of megaevents, with lasting positive and negative effects on the host cities. In recent years, growing evidence of low return on municipal investment, together with increasing public opposition, has contributed to local and national politicians' hesitation to apply for hosting the Olympics (Lauermann, 2022). Of the five bids that cities submitted for the 2024 Summer Olympics, Hamburg withdrew after a referendum, Budapest after a successful citizen petition for a referendum, and Rome after the election of Virginia Raggi from the Five Star Movement as the city's mayor. Paris and Los Angeles were thus left as the only candidates. In 2017, in an unprecedented move, the International Olympic Committee (IOC) announced the hosts of two subsequent Games simultaneously, Paris in 2024 and Los Angeles in 2028. Despite public opposition, not only Olympics but also other megaevents, whether sports events (e.g. football World Cup), World Fairs or other ever larger and more frequent technology and entertainment events continue to proliferate both in the Global North and South. Such events help reshape the host cities and societies (Gardner, 2022; Roche, 2002),

while being embedded in broader global processes and tendencies (Roche, 2017).

Following Müller (2015: 638), this article defines megaevents as temporary events that "attract a large number of visitors, have a large media reach, come with large costs and have large impacts on the built environment and the population". However, what makes megaevents special is not only their size and scale but also their complexity and uniqueness: each event is essentially a first-of-a-kind in its own spatial and temporal context, even when building on experience from earlier times and venues.

The scholarly literature has explored a broad range of economic, social, policy, and planning impacts of Olympics. Economic studies have applied a cost-benefit perspective, concentrating on the evaluation of costs for the construction of the necessary urban infrastructure (the sports facilities, transportation, housing for the athletes), and the operating costs (Baade & Matheson, 2016). Frequently identified problems have included project revisions, cost overruns, delays, and debt (Andreff, 2012). The results from urban geography as well as planning and policy analyses, from a broader spatial and socio-economic perspective, have varied greatly, depending on the focus, assumptions, scoping, framing, and methodologies. Some studies highlight short-term

* Corresponding author.

E-mail addresses: antonia.casellas@uab.cat (A. Casellas), markku.lehtonen@upf.edu (M. Lehtonen).

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benefits such as tourist spending during the Games as well as long-term benefits under the umbrella of “the Olympic legacy” (Chen & Henry, 2020). These include the possible psychological boost to residents and businesses, the improvement of the image of the host city (Kassens-Noor et al., 2019), improved quality of the built environment and transport infrastructure (Kassens-Noor, 2013; Wang & Bao, 2018), and the development of new institutional frameworks for city and metropolitan governance (Geffroy et al., 2021). Other studies have identified an array of negative legacies ranging from severe and nearly chronic budget overruns, which can cripple the host city’s finances for decades (the Montreal Olympics in 1976 generated an estimated shortfall of around \$1.2 billion) to the adoption of neoliberal modes of urban governance, and detrimental changes in planning legislation and urban growth patterns (Chorianopoulos et al., 2010; Hadjimichalis, 2013). Scholars have also drawn attention to problems such as rapid gentrification and displacement of less well-off segments of the population (Gafney, 2016; Lenskyj, 2000; Müller & Gafney, 2018), the exacerbation of existing spatial inequalities, and measures designed to conceal rather than to solve social problems such as homelessness (Chorianopoulos et al., 2010; Cornelissen, 2011; Kennelly, 2015; Lenskyj & Wagg, 2012).

While valuable in and of themselves, these studies of the short- and long-term impacts of Olympics on urban planning, local economy, and social well-being have paid little attention to the broader urban policy and economic dynamics at play in the overall justification, planning, implementation, and evaluation of such megaevents. In this article, we formulate and test a new analytical tool for bridging this gap. We do so by drawing on the lessons and concepts employed by megaproject scholars in the controversy surrounding the so-called Hiding Hand principle advanced by Alfred Hirschman in 1967. In short, Hirschman argued that overoptimism – in essence, underestimation of the implementation challenges ahead – is sometimes necessary to ensure that projects vital for society are undertaken, and to trigger the kind of creativity that underpins societal progress. Hirschman’s idea has been resolutely resisted especially by Bent Flyvbjerg, a leading megaproject scholar. Flyvbjerg (2009; 2011; see also Flyvbjerg & Gardner 2023) sees optimism not as an asset but instead as the main culprit behind the practically chronic failure of megaprojects to reach their objectives. He attributes the failure precisely to overoptimism, that is, either sincere or strategic underrepresentation of the costs and overestimation of the expected benefits by the project proponents.

We illustrate this analytical framework by presenting Olympics as an emblematic example of the diverse roles that optimism plays in the justification, planning, implementation, and evaluation of megaevents. We argue that this novel perspective enables the exploration of the diverse urban policy agendas that are advocated by key stakeholders, and which underpin infrastructure reforms and urban policies associated with megaevents. We focus on two questions central to this approach. First, how does optimism – especially overoptimism in the form of underestimation of the implementation challenges – manifest itself during the justification, planning, implementation, and evaluation of the megaevent? Second, how do the genuine and malevolent types of overoptimism affect the implementation and consequences of the event, at various timescales? We highlight the complexity of the fundamental question of how to measure the success of a megaevent or a megaproject. In particular, we stress the difficulties of incorporating the wider impacts in the evaluation of the performance of such events and projects, given the multiple and irreducible uncertainties. Furthermore, we underscore the unforeseen challenges and the ways in which they were addressed by the planners and organisers of the Olympics. For this analysis, the Barcelona Games serve as a useful illustrative example, which has generated an ample body of scholarship and whose legacies can now be examined with 30 years of hindsight. We illustrate the operation of optimism in four temporal phases: justification, planning, implementation, and *ex post* appraisal.

2. The Hiding Hand controversy

Megaprojects and megaevents are typically characterised by multiple forms of complexity, high economic and political stakes, uniqueness, and considerable economic, environmental, and social impacts that reach well beyond the immediate surroundings of the event or project. In view of the “iron triangle” of (mega)project performance criteria – cost, timetable, and predefined project specifications (e.g., Dimitriou et al., 2017) – megaevents present “pathologies” (Guntton, 2003) similar to those of megaprojects, in that they frequently run over budget and fail to reach their initial objectives. By contrast, unlike megaprojects, a megaevent needs to be delivered on time. However, even this difference must be put into perspective, given that megaevents are often key ingredients in more comprehensive long-term programmes of urban redevelopment. As such, megaevents are subject to similar challenges related to long timescales, including the constant evolution of the context, technologies, institutions, policies, actor alliances, underlying societal values, and objectives. Megaevents are not ephemeral one-off events. Because both their spatial and temporal boundaries reach far beyond the narrowly defined event, also their success should be evaluated through a broader lens.

Planning and *ex ante* assessment of megaevents are vulnerable to the kinds of optimism bias, planning fallacy, and strategic misrepresentation that the mainstream literature highlights as key sources of megaproject “pathologies” (e.g., Flyvbjerg, 2009, 2011, 2014; Guntton, 2003; Priemus, 2010). However, megaprojects and megaevents are organically evolving open systems, whose boundaries are in constant flux (Dimitriou et al., 2017). Hence, the “iron triangle” is an insufficient measuring rod for the appraisal of the success of such events and projects, in terms of the planning, management, implementation and effects at various temporal scales (Lehtonen, 2014). This is particularly significant since megaprojects and megaevents produce “wider” effects, precisely because their very *raison d’être* is often to transform their own context (Vickerman, 2017; Lehtonen, 2019; Ward, 2021). Such projects and events could therefore not be adequately assessed via conventional impact evaluation approaches that require maintaining the contextual variables stable. Moreover, while the organisers retrospectively tend to portray the legacies of megaevents as planned and intended, the outcomes are in fact often highly unpredictable, and result from path dependencies and processes set in motion long before event planning has even started (Gardner, 2022). Finally, the objectives of megaevents and megaprojects change over time, in accordance with changing priorities and values of society. As a result, the “goalposts” of evaluation also change. With the passing of time, the initial objectives of an event may lose much of their relevance as evaluation criteria.

One way to apprehend such wider dynamics – notably the wider unintended, unanticipated, and indirect effects – is Albert Hirschman’s Hiding Hand principle. Hirschman (1967, 13) summarised the principle as follows:

we would not consciously engage upon tasks whose success clearly requires that creativity be forthcoming. Hence, the only way in which we can bring our creative resources fully into play is by misjudging the nature of the task, by presenting it to ourselves as more routine, simple, undemanding of genuine creativity than it will turn out to be.

In other words, Hirschman (1967, 13) argued that, because we constantly underestimate the skill and ability of project managers (public and private promoters in the case of megaevents) to find creative solutions to problems and challenges as they arise, the implementation of major projects requires corresponding underestimation of the challenges ahead. The Hiding Hand entails the idea that practitioners can ultimately “save projects in unforeseen ways during their implementation”, despite – and sometimes precisely because – they ignore or underestimate possible risks, difficulties, and unforeseen contingencies (Ika, 2018: 379). Hirschman’s key concern was creativity, which he saw

as key to societal development. From this perspective, the main virtue of overoptimism in megaproject planning is its capacity to trigger the creative power of project managers, by forcing them to seek innovative solutions to unanticipated problems. Full knowledge of the challenges ahead would discourage project managers from embarking on risky endeavours such as the implementation of a megaproject, and society would thus miss the creativity-generating potential of the megaproject.

Such an account is at odds with the mainstream megaproject literature, which adopts a relatively straightforward linear-rationalist perspective, arguing that successful megaprojects are rare exceptions, not least because of overoptimism and strategic behaviour (Pitsis et al., 2018). Where Hirschman sees optimism as (sometimes) essential for success, the mainstream megaproject scholarship, represented notably by Flyvbjerg (2014; 2017), describes the recurrent megaproject failures as outcomes of two features. The first, according to Flyvbjerg dominant for projects that do not entail major political controversy, relates to cognitive biases – the inherent optimism of human beings (“optimism bias”, or “planning fallacy”). The second, “strategic misrepresentation”, which van Marrewijk et al. (2008) have dubbed as “malevolent design”, would prevail in politically highly charged situations that arguably characterise most megaprojects and megaevents. In such a case, project planners and proponents would intentionally overestimate the benefits and underestimate the costs and uncertainties to ensure that their project gets implemented. Flyvbjerg (2016; 2018) has blamed Hirschman for biased data collection (focussing on unexpected successes), a very small sample size of eleven cases, and the presumably unfounded claim that Hiding Hand would constitute a general principle of action. Flyvbjerg refers to his own analysis of more than 2000 cases, of which 80 % displayed significant cost overruns and benefit shortfalls, and argues that a ‘Malevolent Hand’, resulting from overoptimism and strategic misrepresentation, was in operation in most cases. Flyvbjerg and Gardner (2023) also contest the idea that creativity would emerge under external pressure, and argue instead that to flourish, creativity requires environments shielded from outside pressures. Flyvbjerg and Gardner recognise the usefulness of optimism and creativity, but only at the phase of careful project planning, not implementation, which should adhere to standardised procedures and include continuous control and monitoring (ibid.).

The basic dilemma of optimism has also been addressed in the literature on techno-scientific promises and innovation. The proponents of an innovation need to display a certain degree of optimism and present bold promises, to engender the enthusiasm, courage, and optimism needed to turn the innovation into a concrete application. Optimism and enthusiasm are necessary among the project managers and the broad range of actors involved, including decision-makers, investors, and the public at large. However, this very optimism – the necessary bold promises – risks leading to disappointment, and often underlies the recurrently observed budget and schedule overruns, and benefit shortfalls (e.g., Joly, 2010).

Importantly for the purposes of this article, the two “hands” highlight the distinction between “genuine” and “strategic” optimism: while the Hiding Hand describes situations in which the optimism of the managers is sincere in that it is based on true lack of knowledge, the Malevolent Hand represents strategic use of overoptimism and “intentional gaming” for private benefit rather than general interest. Yet, even this distinction deserves being nuanced given that strategic overoptimism can entail “virtuous lying” and be motivated by sincere concern for the common good, founded on a belief that the event/project in question is in the wider public interest. Such a belief relies on the notion that “we know better” – that it is the duty of the expert to do what s/he “knows” is in the greater interest of society, even if doing so would require misrepresenting the facts or outright lying. Therefore, distinguishing between sincere and strategic overoptimism is not only empirically challenging but may have limited relevance from the perspective of megaevent legacies: whether sincere or strategic, overoptimism produces tangible consequences. Table 1 summarises the key elements of

Table 1
The Hiding and the Malevolent hands.

	Hiding Hand	Malevolent Hand
Role and nature of optimism	• Underestimation of the challenges ahead • Overestimation of one’s own capacity to overcome challenges	• Overoptimism as a fundamental human trait; “planning fallacy” • Overoptimism as a strategy: strategic misrepresentation • Optimism and creativity adequate elements of careful project planning – but should have no place in implementation
Outcomes of (over) optimism	• Creativity: innovative solutions, adaptation to unexpected circumstances • Projects useful for society get implemented (sometimes) • Possibility of short-term management failures to produce long-term societal successes	• Chronic failure in view of the iron triangle of project performance criteria (cost, schedule & predefined project objectives and specifications) • Survival of the unfittest (the most overoptimistic projects get funded) • Short-term management failures = waste of society’s resources also in the long run

Source: authors.

the Hiding and Malevolent hands.

3. Data and methods: the Hiding Hand controversy as a tool for analysing megaevents

This article illustrates the potential of the Hiding Hand controversy as a theoretical lens through which to investigate the dynamics of planning, implementation, and impacts of megaevents. We argue that exploring the extent to which a Hiding or a Malevolent Hand operated at different stages of the megaevent can provide vital insights into the multiple roles of optimism in megaevent policy and management. Ultimately, it can help to inform the debates on the wider impacts and their relationship with the stated, official objectives of the megaevent in question. The operation of the Hiding Hand and the Malevolent Hand can usefully be analysed at four phases of a megaevent: justification, preparation, implementation, and appraisal (Table 2).

Focusing on the identification of when, how, and why overoptimism emerged, the description of the justification phase provides an inductive lens through which to analyse the political and economic context in which the urban megaevent took place and evolved. The analysis of the operation of optimism in the planning phase helps to identify the urban

Table 2
The Hiding Hand controversy as an analytical tool.

Megaevent Phases	Questions
Justification	What were the predicted outcomes of hosting the event and who were expected to benefit? Were the justifications overly optimistic? Was the optimism sincere or strategic?
Planning	What were the declared and underlying urban infrastructure and planning objectives? Were the planning objectives overly optimistic? Was the optimism sincere or strategic?
Implementation	What were the implementation challenges? Did the challenges result from (over)optimism? Was the optimism sincere or strategic? What creative solutions were generated to address the challenges?
Evaluation	What were the short-, medium-, and long-term outcomes and consequences of the event? To what extent can the outcomes be attributed to the operation of the Hiding and the Malevolent Hand?

Source: authors.

policy and planning goals that the main stakeholders expected the Olympics to reach. The identification of optimism in the implementation phase allows the exploration of the ways in which the project organisers coped with unanticipated challenges, that is, how they deployed creativity to address the unforeseen problems that arose during the implementation. Finally, the section dedicated to the *ex post* evaluation phase explores the short-, medium-, and long-term outcomes of the Games as an organically evolving open system, with a focus on the “wider benefits” and “wider damage”.

The case of Barcelona 1992 Olympics serves as an example that allows us to illustrate the potential of the Hiding Hand controversy as an analytical tool. This article does not present a detailed empirical case study of the Barcelona Games but harnesses instead the Barcelona example to the purposes of theory-development, by suggesting a four-phase analytical framework applicable to megaevents more generally.

The key data for the study consisted of a diversity of documentation ranging from scholarly publications to policy and planning documents, which provided information on the political and social context in which the Games were organised. Evidence was gathered through qualitative analysis of archival and media sources, extensive fieldwork on the urban planning policies, note-taking during direct observation of public meetings, and formal and informal interviews with key stakeholders and decision-makers between 1985 and 2010. A large part of the empirical material was collected as part of the PhD thesis work of Casellas (2003), complemented by further analysis of the urban evolution of Barcelona in the years following the Games (Casellas, 2006; 2007; 2011; 2016; Casellas et al., 2010). The analysis of optimism involved deep data immersion. Data crystallisation was then partly guided by the Hiding Hand controversy, which emerged as a reflexive theme (Braun & Clarke, 2019), as an interpretative story from the data, produced at the intersection of our theoretical tool, our analytical resources and skills, and the empirical data. Lacunae in the existing data were filled, as needed, via focused search for new empirical data.

In the following section, we test the potential of the Hiding Hand controversy in four lifecycle phases of a megaevent. The last section of the article discusses whether the optimism observed in the various phases of the organisation of the Barcelona Games represented the operation of a Hiding Hand, generating creativity and “wider benefits” that would otherwise not have been mobilised, or whether instead overoptimism led to short-term event management failures and longer-term “wider damages”. We conclude by discussing the potential usefulness of the analytical framework of Hiding Hand vs. Malevolent Hand for the analysis of megaevents more broadly.

4. Analysis: testing the Hiding Hand controversy as an analytical tool

4.1. Overall urban upgrade as an optimistic justification for the Games

In Barcelona, local politicians were leading promoters of the Games, and showed high expectations concerning their benefits for the city. After decades of urban policy neglect during Franco’s dictatorship (Wynn, 1979), in the early 1980s the first democratically elected socialist local government saw hosting the Olympics as a pretext for an ambitious overall city upgrade (Balibrea, 2001; De Balanzó & Rodríguez-Planas, 2018; Degen & García, 2012; Moragas and de Moragas & Botella, 1995). At that time, the city faced severe challenges, which included increasing population density, degradation of the urban landscape, lack of open spaces, expansion of peripheral poor neighbourhoods, a dramatic lack of financial resources, and mounting public debt (Marshall, 2004).

The bid book, presented to the IOC on March 1, 1986, captured the enthusiasm and the resources deployed by the promoters. The documentation – the result of work by 750 experts over a period of four years – included seven volumes that amounted to 1400 pages, 3000 photographs, 2000 maps, and 50 h of audio and video material. Mayor

Maragall, the key political leader at the time (Mauri and Uría, 1998), asserted in the introduction of a book explaining the bid: “I have no doubt that the Olympic Games (...) will help us to create, as the bid has done so far, a more cohesive and healthier society (COOB, 1987: 7).

The urban policies in Barcelona had prioritised the recovery of downtown and peripheral working-class neighbourhoods via the deployment of a pragmatic method of executing modest urban regeneration projects. Most of the projects were implemented on public land, more specifically, in public spaces such as neighbourhood squares (Bohigas, 1999). In this context, the Olympics served as a tool that would justify the national and regional funding sorely needed to transform the city. The objectives that Mayor Maragall presented in 1988 in the book *Urbanism in Barcelona: Plans towards 1992* illustrate the extent of overoptimism concerning the potential of the Games to engender an overall urban remake. The overall objectives were: 1) renewing the historic centre; 2) upgrading the peripheral neighbourhoods; 3) improving the road network; 4) opening the city to the sea; 5) constructing the sport facilities, hotels, and housing needed for the Olympics; and 6) creating new economic centres in the city to attract businesses and to improve the quality of life of city dwellers (Maragall, 1988).

Due to a dramatic lack of financial and administrative resources, the radical urban makeover, with the Games as a pretext, was the cornerstone of the kind of overoptimism described by the Hiding Hand principle: *a misjudgement of the nature and challenges of the task ahead*. Sincere overoptimism probably combined with strategic considerations along the lines of the Malevolent Hand yet underpinned by the desire to “do good”: the promoters saw the Games as a strategic tool to improve the infrastructure and urban built environment and, as a result, to enhance the quality of life of a large number of citizens expected to benefit from the urban remake.

4.2. Ex ante planning: sincere overoptimism concerning the support for the Games

Seeking to ensure an equal distribution of urban renewal projects across the city, planners designated four areas to develop, in preparation for the Games. The first was Vall d’Hebron, a working-class neighbourhood that since its development in the 1960s had remained physically and socially cut off from the city. The second was the Montjuïc hill, an area redeveloped to host the 1929 Universal Exposition. Montjuïc was one of the few open spaces in Barcelona, but had remained in a poor state due to neglect. The third area was on a coastal strip of industrial land, where the Olympic village was to be constructed. The fourth area was at one of the entrances to Barcelona, hosting diverse university facilities (COOB’92; Martorell et al., 1992; Moix, 1994). To mobilise the broad range of resources (administrative, organisational, knowledge, finance, and so on) needed, the public sector had to build alliances with business, and eventually with the overall population.

Fieldwork undertaken at that time illustrates the genuine overoptimism of the promoters regarding the level of trust and support from the city’s economic elites. After Barcelona had won the Olympic bid, a businessman interviewed regarding the expectations on the Games asserted that “as you may realise, a macro-event organised by a socialist (indirectly referring to Mayor Pasqual Maragall, who was a member of the Catalan Socialist Party) and a communist (referring to Josep Maria Abad, the director of the Barcelona Olympic Organising Committee COOB’92, and member of the Catalan communist party) will never work”. The powerful local hotel corporations’ reticence to invest in the construction of new hotels further shows how the organisers failed to anticipate the lack of support from the local business elite.

The analysis of grassroots urban protests during the planning phase provides further evidence of the presence of the Hiding Hand. The scepticism and opposition from grassroots movements were an unexpected challenge for the organisers as illustrated by their reaction to the 1987 downtown protest campaign “There is hunger here! Campaign for

food solidarity". In their manifesto, the protesters asserted that in "the successful, technological, Olympic Barcelona" there were people who did not have adequate access to a basic right: food. It stressed that hunger was only the most visible symptom of deeper problems, which included the deplorable living conditions of the unemployed in the downtown Barcelona, and poor access to basic health, education, and cultural services. The manifesto underscored that antisocial conduct such as delinquency was the result of neglect by the political establishment and concluded that the campaign sought not only to collect food donations from charitable donors but also to stimulate initiatives of mutual civic support (Comellas, 1995). It challenged the municipal and Catalan governments for their inability to address the problem of social marginality and their "generally unsupportive and defensive behaviour". Lacking adequate knowledge and understanding of the prevailing social situation, the local and regional governments reacted by blaming each other for having organised the downtown protest campaign. Once it became clear that the campaign was propelled by a grassroots movement without political affiliation, the political debate turned into a confrontation whereby each party sought to discredit each other. This revealed the tension between the two leading political parties that, at the time, were forced to cooperate in the organisation of the Games, but also shows the inability of the organisers to estimate the extent of the challenges. To deal with these challenges, the organisers of the Games had to employ innovative urban planning, policy, and governance solutions.

4.3. Creative implementation solutions to underestimated and unexpected challenges: The hiding and the malevolent hands in operation

Because of the discrepancy between the scale of ambition of the urban transformation goals on the one hand, and the weak economic, political, and social support and resources on the other, the city council faced numerous unforeseen challenges during the implementation phase. As described in the previous sections, overoptimism led to four types of challenges, which called for innovative solutions. First, politically, the conflicts and tensions between the city council and the Catalan government were compounded by the interest of the Spanish government in capitalising on the event, by harnessing the Games for its own political purposes (McNeill, 2005). Second, from the private-sector perspective, the organisers had to cope with the scepticism and reluctance of the local business elites to engage in support of the Games. Third, socio-economically, there was mistrust not only among the local economic elites but also among diverse grassroots community organisations towards the Games and their organisers. Finally, from a technical-economic perspective, the need to construct the sports facilities and other infrastructure as required by the IOC put a strain on public resources.

Chance played an unexpected key role in mitigating the political tensions. The inauguration of the Olympic Stadium at the opening ceremony of the 5th IAAF Athletics World Cup in September 1989 turned into a fiasco, for a combination of unanticipated troubles: 1) heavy rain – a rare phenomenon in Barcelona – flooded the stands of the new Olympic stadium, thus exposing the inefficiency of the drainage system; 2) King Juan Carlos I arrived over half an hour late; 3) the withdrawal of the only private sponsor forced the organisers to simplify the ceremony; 4) the crowd whistled at the Spanish national anthem and at the King; and 5) in protest, a left-wing Catalan independence group launched rocket flares over the stadium, whereas the conservative independence group showed banners with the slogan "Freedom for Catalonia" (Badia et al., 2017; El País, 1989; McNeill, 2005). Less than three years before the Games, and facing the prospect of a total humiliation, the political entities involved understood that they had to cooperate and coordinate their efforts. Estimates of the financial contribution from the Catalan government range from 22 % (McNeill, 2005) to 33.3 % (Raventós, 2000) of the total public investment. This contribution mainly served to finance sports and road infrastructure and the development of Olympic

sites outside the city. In the city, the Spanish state retained the overall control by contributing 51 % of the total capital, while the remaining 49 % were provided by the Barcelona City Council (Brunet, 2002).

From an economic perspective, strategic planning turned out to be a key innovation introduced by the public sector to establish a dialogue with the private sector. Initially, the city council did not consider it necessary to include labour unions in the executive commission, and primarily sought to gain the trust of the private sector only. However, following a seminar held in July 1988 to draw lessons from the experiences of cities such as Birmingham, Milan, Rotterdam and San Francisco, Mayor Maragall realised it would be important to invite to the Executive Commission of the Strategic Plan also representatives of the two main trade unions of the city, the Comisiones Obreras (Workers' Commission) and the Unión General de Trabajadores (General Workers' Union) (Raventós, 1998). Even though the drawing up of the plan was open to participation by community groups, and community participation was actively sought, ultimately the plan was written by a technical committee composed of 526 members, mainly academics and representatives of Barcelona's economic and institutional interests. The plan was presented to various publics at conferences and seminars. Public and private entities sponsored an exhibition that attracted 95,000 visitors, of whom about 4000 provided feedback in writing (Marshall, 1990). As shown by our interviews with decision-makers, Barcelona's Social and Economic Strategic Plan served as a key urban planning and policy tool for building consensus between the socialist council and the local business elite. This would help create a shared vision and allow the city to capitalise on the growth dynamics following from the infrastructure investment and the international visibility that the Games were expected to generate (Raventós, 1998; 2000).

From a social perspective, the unprecedented mobilisation of volunteers served as an innovative measure to summon much needed public support. Andreu Clapés (1992), the Assistant Director of the organisation charged for managing the participation of volunteers, estimated that of the more than 43,000 of the citizens that volunteered, 3/4 were less than 23 years old. The organisation of the Games mobilised around 90,000 people, of which approximately 35,000 were volunteers (Brunet, 1995). This proved to be an unparalleled and effective means of building support and goodwill for the Games, especially among young people.

Yet another major unexpected problem that required creativity and innovative solutions related to the hosting of visitors and country delegations. At that time Barcelona was not a major tourist destination and was extremely poorly equipped to organise a megaevent as demanding as the Olympics. To incentivise the private sector, the city council approved a hotel development plan. A zoning reform allowed the owners of the urban land previously designated for public services and recreational facilities to present plans for building luxury hotels. The final agreement granted such hotel developers an administrative concession for 50 years on publicly owned land (Bramona, 2001). Nevertheless, by the time of the Games, of the eight luxury hotels approved in 1989 only six had been finished. As a radical and innovative solution, the organisers decided to bring cruise ships to the port and use these to provide an extra 11,000 luxury beds (Riding, 1992).

If the above-described innovative solutions can be seen as examples of the Hiding Hand, the creative response of the local organisers to the major technical-economic challenge – the requirements of the IOC regarding sports facilities – rather exemplified the Flyvbjergian Malevolent Hand. The 1988 Seoul Olympic Stadium had a capacity of 100,000 seats, while the Atlanta 1996 stadium could host 85,000. Concerned with high construction costs and the risk of having to manage an oversized infrastructure after the Games, the Barcelona Olympic organising committee opted for a smaller stadium of only 60,000 seats, well below the minimum of 70,000 required by the IOC. To provide the required additional 10,000 seats, the organisers installed temporary seats in corridors and stairs, and built a temporary extension. All these were removed after the Games, although the IOC had rejected this option in

1990 (El País, 1990). The solution was ultimately implemented because, as one of the key decision-makers admitted in an interview, the COOB'92 repeatedly misled the IOC by reassuring it that once finished, the upgraded stadium, originally built in 1939, would comply with all IOC requirements.

4.4. *Ex post appraisal: the hiding and malevolent hands from a short-, medium- and long-term perspective*

A broad consensus prevailed soon after the Olympics that the Games had helped to put Barcelona on the world map (Kennett & Moragas, 2006). An opinion survey among the Barcelona residents on the success of the Olympics gave an average mark of 8.70 out of 10. Also, a visitor survey revealed a high level of satisfaction, especially with regard to the quality of the events, atmosphere, and facilities (Brunet, 1995).

In the short and medium term, the 300 infrastructure projects spurred by the Games, including the construction of new roads, the airport renovation, and a vast improvement of the telecommunications, gave the city a boost and improved the quality of life of its residents. The creation of new public spaces, sports facilities, and the opening of the city to the sea further contributed to the improvement. From this perspective, the various ways in which optimism operated in the organisation of the Games served a useful purpose, along the lines of the Hiding Hand.

Evaluating the Barcelona Games from a 30-year perspective, as an organically evolving open system, shows a more complex picture, which lends support to both the Hiding and the Malevolent Hand hypotheses. A lasting outcome of the Olympics was the consolidation of the novel public-private cooperation and strategic planning arrangements that would continue to shape the urban agenda during the subsequent three decades. The need of the public sector to mobilise a broad range of resources (administrative, organisational, knowledge, financial, and so on) to enable the organisation of the Games forced the local government to seek private-sector allies which, over the subsequent three decades, gained increasing influence in Barcelona's urban development policies. The initial public-private partnership model generated by the Olympics to secure the construction of the Olympic Village was subsequently extended to downtown redevelopment. The private sector, which provided much-needed economic resources, over time came to shape the Barcelona urban redevelopment agenda to an ever greater degree. Over the subsequent years, the public-private partnership generated by the Olympics progressively shifted risks towards the public sector (Sabaté Bel & Tironi Rodó, 2008).

Initially introduced as an innovative policy tool designed to achieve consensus on the direction of urban growth after the Games, strategic planning quickly evolved into an instrument designed to facilitate the approval and implementation of territorial, sectorial, social, and economic development plans and urban renewal projects in the city, and later in the entire metropolitan area. The first strategic city plan was approved in 1990, the second in 1994, and the third in 1999. Two metropolitan strategic plans followed, in 2003 and 2010. Many other sectoral strategic plans – culture, sports, tourism, etc. – were adopted. Strategic planning functioned as a mechanism used by the local administration to promote the inclusion of a wide spectrum of economic and social agents in political decision-making. However, in this process, the local authorities gave priority to achieving consensus among actors that possessed valuable economic, financial, and technical resources (Capel, 2005; Casellas, 2016). The mottos of the successive strategic plans reflect this evolution: “Put Barcelona on the map” in 1990, “consolidate international visibility” in 1994, “shifting toward the knowledge economy” in 1999 “focusing on the metropolitan scale” in 2003, and finally, “making Barcelona and its metropolitan region attractive to business and the creative class” in 2010 (Estela, 2018).

Many international city rankings from the past decade have placed Barcelona as among the top cities in Europe the quality of life. The city has also become a leading world urban tourist and cruise destination

(Garay et al., 2014; Vayá et al., 2018). Furthermore, Barcelona has sought to market itself as an advanced knowledge economy (Pareja-Eastaway & Pique, 2011) and a smart city (Bakıcı et al., 2013; Bibri & Krogstie, 2020). Yet, like many other cities that attract tourists and speculative capital, it suffers from gentrification (Charnock et al., 2014; Cocola-Gant & Lopez-Gay, 2020), housing shortages (Elorrieta et al., 2022), air and noise pollution (Lagonigro et al., 2018; Perdiguerro & Sanz, 2020), and other downsides of mass tourism (Cocola-Gant et al., 2020; Hughes, 2018; Milano et al., 2019). Although it is impossible to establish a direct causality between the 1992 Olympics and these outcomes, the Olympics indisputably constituted a turning point for Barcelona, propelling a shift in urban planning towards a model in which the private sector interests play an increasingly dominant role. This trend continued at least until 2015 when a new left-wing political coalition took power in the city council.

5. Discussion

The analysis of the Barcelona case shows that the organisation of the Games was characterised by plenty of improvisation, experimentation, and learning by doing, combined with a good dose of luck. Focusing on (over)optimism, we identified four unforeseen challenges that the organisers faced in the planning and implementation phases, namely: a) the lack of support from the local business elite, b) the lack of support from community groups, c) constraints imposed by the requirements of the IOC, and d) the lack of cooperation between the local, regional, and national governments and administrations. In the face of such challenges, the genuine overoptimism was accompanied by skilful mobilisation of creative resources and narratives designed to muster support for the Games and conceal some of their downsides.

Overoptimism served two main purposes, in line with Hirschman's Hiding Hand proposition. First, it fostered creativity, and second, it helped to overcome the obstacles to the implementation of what the promoters of the Games believed was a transformative megaevent, vital for the medium-to-long-term development of the city. In many respects, the Barcelona Games exemplified the operation of the Hiding Hand, whereby both sincere and strategic overoptimism serve noble purposes rather than individual interests. Moreover, the formation of the public-private alliances in Barcelona facilitated “getting things done” – mentioned in the literature as one of the positive aspects of the Malevolent Hand (Anheier, 2016). Nevertheless, the collaborative relationships needed to get the Games up and running were not cost-free but in the long run shaped the urban policies, influenced the speed of the evolution of the city, and shifted to private actors some of the power hitherto exercised by the public sector.

Overall, the justification, planning, implementation, and evaluation phases of the Barcelona Games were characterised by sincere and strategic overoptimism, unforeseen challenges, and innovative solutions, including the strategic provision of misleading information. At times, the planners and promoters of the event lacked the knowledge necessary to truly appreciate the many challenges ahead. Yet on other occasions they intentionally overestimated the benefits and underplayed the risks in order to gather political support, enrol key agents such as investors, and in this way enable the implementation of the urban redevelopment and institutional arrangements they considered vital for the future of the city.

The methodological approach developed in this article exemplifies the context-dependence of the very definitions of success and failure, and thus highlights the limitations of efforts to simply transfer best practices from one city or megaevent to another. To the extent that the Barcelona Games were a success, they were so largely as an outcome of fortunate combination of politico-economic circumstances: transition to a democracy, pre-existing plans for urban renewal, (relative) consensus on the need and direction of change, and previous involvement of leading politicians in grassroots activism. However, the example also showed that to turn fortunate circumstances into concrete success, both

sincere and strategic uses of optimism were fundamental. As the Hiding Hand principle suggests, optimism triggered creativity and generated a virtuous circle of unexpected challenges, creative solutions, and learning.

This study has showed that the Hiding Hand controversy can inform the analysis of megaevents more broadly, in at least four ways. First, the identification of optimism among the promoters in relation to the foreseen outcomes helps to unveil the dominant urban agendas motivating the idea of hosting the megaevent. In the case of Barcelona, unlike in many other megaevents, local politicians, in coalition with technocrats, used the Olympics as a pretext for an overall makeover of the city. This approach shaped the entire process. Second, in the planning phase, the identification of optimism and unforeseen challenges draws attention to the unavoidably contested nature of megaevents and helps to explore the plural agendas of the key agents, the power relations between them, and the various forms through which power is exercised in support of or against the megaevent. In Barcelona, the optimism of the public-sector organisers at the planning phase revealed a reality that was far from a unity and a consensus, and laid the basis for four types of creativity-generating challenges in the implementation phase. Third, in the implementation phase, the examination of the challenges and the creative response measures deployed helps to understand the operation of optimism within its local context, thereby allowing the development of “middle-range theories” concerning the context-mechanism-outcome configurations in the face of uncertainty. Fourth, in the evaluation of the short-, medium-, and long-term outcomes and consequences of overoptimism and mobilisation of creativity, the criterion of success needs to be considered in the light of two elements. First, the very long timescales involved mean that societal goals, objectives, and values evolve over the project’s lifetime, and therefore a short-term project management failure can, in the long term, turn out to be a success, and vice versa: a project that meets its “iron triangle” project management objectives can, over time, turn out to be useless or even harmful for society. Second, and following on from the above, success and failure are multifaced, contextual, and dependent on the perspective, notably the definition of project boundaries. Megaevent evaluation should therefore cast a wider net and explore the consequences of such events as part of the broader policies, programmes, and strategies. These include the long-term objectives, dynamics, and politics of urban planning, within their national and often even international context.

For the megaproject literature, the Barcelona case demonstrates that sincere and strategic overoptimism operated in tandem, and that the distinction between the “Hands” is not necessarily the most important aspect. Examples such as the way in which COOB’92 misled the IOC regarding the capacity of the Olympic stadium show that even strategic misrepresentation and the Malevolent Hand can serve useful purposes. It would not only be naïve and unrealistic to assume that the project promoters have only noble intentions, but such noble intentions are not always necessary and by no means a guarantee of success. Whether optimism serves socially beneficial purposes and how depends on broader configurations of the context, the measures employed, and the forms of optimism, not on whether the optimism is sincere or strategic. Instead, these types of optimism, manifested in the Hiding Hand and the Malevolent Hand, must be examined within a broader context and perspective of historical, political, institutional trends, as well as local and temporal contingencies.

6. Final remarks

This article outlined an analytical approach to the study of megaevents, which is novel in three main ways. First, the controversy over the Hiding Hand principle can help unveil key urban policy agendas and strategies underpinning the hosting of megaevents. Second, the identification of optimism – whether it be sincere or strategic – can help better understand the power dynamics among different stakeholders that are a vital part of urban planning processes in general, and the decision-

making around megaevents in particular. Third, the study of optimism allows megaevents to be examined within their specific urban contexts and cautions against simplistic notions of “best practices” straightforwardly transferable from one context to another. In our case study, even though there is a wealth of literature portraying the Barcelona Olympics as a successful megaevent and as an example for other host cities to follow, the analysis from the perspective of the Hiding Hand controversy highlights the dependence of megaproject success on complex and context-specific idiosyncrasies.

The Barcelona case served here primarily an illustrative purpose – as a means of demonstrating the potential of the Hiding Hand controversy as a theoretical approach. Future research could delve deeper into the case-specificities, for example by analysing the nature of optimism and overoptimism at a specific stage of a megaevent – in its justification, planning, implementation, and evaluation. Alternatively, useful insights could be gained via comparative analysis of various megaevents using the Hiding Hand controversy as the analytical lens.

CRedit authorship contribution statement

Antònia Casellas: Conceptualization, Data curation, Formal analysis, Methodology, Writing – original draft, Writing – review & editing. **Markku Lehtonen:** Conceptualization, Data curation, Formal analysis, Methodology, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

- Andreff, W. (2012). The winner’s curse: Why is the cost of sports megaevents so often underestimated? In W. Maennig, & A. S. Zimbalist (Eds.), *International handbook on the economics of mega sporting events* (pp. 37–69). Edward Elgar Publishing.
- Anheier, H. K. (2016). Of hiding hands and other ways of coping with uncertainty: A commentary. *Social Research*, 83(4), 1005–1010.
- Baade, R. A., & Matheson, V. A. (2016). Going for the gold: The economics of the Olympics. *Journal of Economic Perspectives*, 30(2), 201–218.
- Badia, J., Bellmunt, J., Claret, J., Fuster-Sobrepere, J., Nel-lo, O., & Brugué, Q. (2017). *Pasqual Maragall. Pensamiento y acción*. RBA Libros.
- Bakici, T., Almirall, E., & Wareham, J. (2013). A smart city initiative: The case of Barcelona. *Journal of the Knowledge Economy*, 4, 135–148.
- Balibrea, M. (2001). Urbanism, culture and the post-industrial city: Challenging the “Barcelona Model”. *Journal of Spanish Cultural Studies*, 2(2), 187–210.
- Bibri, S. E., & Krogstie, J. (2020). The emerging data-driven Smart City and its innovative applied solutions for sustainability: The cases of London and Barcelona. *Energy Informatics*, 3, 1–42.
- Bohigás, O. (1999). Valorización de la periferia y recuperación del centro. In P. Maragall (Ed.), *Europa Próxima. Europa, Regiones y Ciudades* (pp. 199–214). Barcelona: Edicions Universitat de Barcelona.
- Bramona, A. (2001). General secretary Barcelona’s hotel association. *Interviews*. May 3.
- Braun, V., & Clarke, V. (2019). Reflecting on reflexive thematic analysis. *Qualitative research in sport, exercise and health*, 11(4), 589–597.
- Brunet, F. (1995). An economic analysis of the Barcelona’92 Olympic games: Resources, financing and impact. In d. M. Moragas, & M. Botella (Eds.), *The keys to success: The social, sporting, economic and communications impact of Barcelona’92* (pp. 203–237). Barcelona: Servei de Publicacions de la UAB.
- Brunet, F. (2002). Anàlisi de l’impacte econòmic dels Jocs Olímpics de Barcelona, 1986–2004. In *Barcelona: l’herència dels Jocs (1992–2002)* (pp. 245–274).
- Capel, H. (2005). *El modelo Barcelona: un examen crítico*. Barcelona: Ediciones del Serbal.

- Casellas, A. (2003). *The Barcelona model? Agents, policies and planning dynamics in tourism development*. Ph.D Thesis. Department of Urban Planning and Policy Development. Rutgers, the State University of New Jersey.
- Casellas, A. (2006). Las limitaciones del "modelo Barcelona": Una lectura desde Urban Regime Analysis. *Documents d'Anàlisi Geogràfica*, 48, 61–81.
- Casellas, A. (2007). Gobernabilidad, participación ciudadana y crecimiento económico: adaptaciones locales a estrategias globales. *Scripta Nova*, 11(243), 10.
- Casellas, A. (2011). Governance and city. The evolution of public-private partnership model in Barcelona. In J. M. Montaner, F. Álvarez, & Z. Muxí (Eds.), *Critical Files. The Barcelona Model, 1973-2004* (pp. 62–66). Barcelona: Ajuntament de Barcelona, Departamento de Composición Arquitectónica del ETSAB-UPC.
- Casellas, A. (2016). Desarrollo urbano, coaliciones de poder y participación ciudadana en Barcelona: una narrativa desde la geografía crítica. *Boletín de la Asociación de Geógrafos Españoles*, 70, 57–75.
- Casellas, A., Dot, E., & Pallares-Barbera, M. (2010). Creación de imagen, visibilidad y turismo como estrategias de crecimiento económico de la ciudad. *Finisterre. Revista Portuguesa de Geografía*, Vol. XLV, n.º 90, 153–172.
- Charnock, G., Purcell, T. F., & Ribera-Fumaz, R. (2014). City of rents: The limits to the Barcelona model of urban competitiveness. *International Journal of Urban and Regional Research*, 38(1), 198–217.
- Chen, S., & Henry, I. (2020). Assessing Olympic legacy claims: Evaluating explanations of causal mechanisms and policy outcomes. *Evaluation*, 26(3), 275–295.
- Chorianopoulos, I., Pagonis, T., Koukoulas, S., & Drymoniti, S. (2010). Planning, competitiveness and sprawl in the Mediterranean city: The case of Athens. *Cities*, 27(4), 249–259.
- Clapés, A. (1992). The Olympic volunteers and the challenge of 92. *Catalonia*, 22–23.
- Cocola-Gant, A., Gago, A., & Jover, J. (2020). Tourism, gentrification and neighbourhood change: An analytical framework-reflections from southern European cities. In *The overtourism debate* (pp. 121–135). Emerald Publishing Limited.
- Cocola-Gant, A., & Lopez-Gay, A. (2020). Transnational gentrification, tourism and the formation of 'foreign only' enclaves in Barcelona. *Urban Studies*, 57(15), 3025–3043.
- Comellas, J. (1995). *Aquí hi ha Ganar! Debat Sobre la Marginació Social a Barcelona*. Barcelona: Editorial Afers.
- COOB'92. (1987). *Com hem guanyat els Jocs Olímpics: llibre-catàleg de la Mostra sobre la història de la candidatura*. Barcelona: COOB'92.
- COOB'92. (1992). *Memòria oficial dels Jocs de la XXVa Olimpíada: Barcelona 1992*. V.3. L'organització: la preparació dels Jocs. Barcelona: COOB'92. <https://www.barcelonaolimpica.net/ca/b/memoria-oficial-jocs-olimpics-barcelona-1992-3/>.
- Cornelissen, S. (2011). Mega event securitisation in a third world setting: Global Processes and Ramifications during the 2010 FIFA World Cup. *Urban Studies*, 48(15), 3221–3240.
- De Balanzó, R., & Rodríguez-Planas, N. (2018). Crisis and reorganization in urban dynamics. *Ecology and Society*, 23(4). <https://doi.org/10.5751/ES-10396-230406>
- de Moragas, M., & Botella, M. (Eds.). (1995). *Les Claus de l'èxit: impactes socials, esportius, econòmics i comunicatius de Barcelona '92*. Bellaterra: Centre d'Estudis Olímpics, Universitat Autònoma de Barcelona.
- Degen, M., & García, M. (2012). The transformation of the 'Barcelona model': An analysis of culture, urban regeneration and governance. *International Journal of Urban and Regional Research*, 36(5), 1022–1038.
- Dimitriou, H. T., Ward, J. E., & Wright, P. G. (2017). Mega projects and mega risks: Lessons for decision-makers of large-scale transport projects: OMEGA Centre lessons derived from European, US and Asia-Pacific case studies. In M. Lehtonen, P.-B. Joly, & L. Aparicio (Eds.), *Socioeconomic evaluation of megaprojects: Dealing with uncertainties* (pp. 62–79). Routledge.
- El País. (1989). La inauguración del estadio fue boicoteada por las juventudes de Convergència y la Crida. https://elpais.com/diario/1989/09/09/deportes/621295210_850215.html (Retrieved: May 13, 2023).
- El País. (1990). El COOB quiere una normativa singular para ampliar Montjuïc. https://elpais.com/diario/1990/02/20/deportes/635468406_850215.html (Retrieved: January 21, 2023).
- Elorrieta, B., Cerdan Schwitzguébel, A., & Torres-Delgado, A. (2022). From success to unrest: The social impacts of tourism in Barcelona. *International Journal of Tourism Cities*, 8(3), 675–702.
- Estela, O. (2018). La planificació estratègica metropolitana a Barcelona: cap a una tercera generació de plans?. In Vol. 1, Núm. 61. *Papers: Regió Metropolitana de Barcelona: Territori, estratègies, planejament, [en línia]*, 2018 (pp. 42–51).
- Flyvbjerg, B. (2009). Survival of the unfittest: Why the worst infrastructure gets built – And what we can do about it. *Oxford Review of Economic Policy*, 25(3), 344–367.
- Flyvbjerg, B. (2011). Over budget, over time, over and over again: Managing major projects. In P. W. G. Morris, J. K. Pinto, & J. Söderlund (Eds.), *The Oxford Handbook of Project Management* (pp. 321–344). Oxford: Oxford University Press.
- Flyvbjerg, B. (2014). What you should know about megaprojects and why: An overview. *Project Management Journal*, 45(2), 6–19.
- Flyvbjerg, B., & Gardner, D. (2023). *How big things get done: The surprising factors behind every successful project, from home renovations to space exploration*. New York: Penguin.
- Gafney, C. (2016). Gentrifications in pre-Olympic Rio de Janeiro. *Urban Geography*, 37(8), 1132–1153.
- Garay, L. A., Cànoves, G., & Prat, J. M. (2014). Barcelona, a leader destination in cruise-passenger tourism: Keys, impacts and facts. *International Journal of Tourism Sciences*, 14(1), 23–49.
- Gardner, J. (2022). *A contemporary archaeology of London's mega events: From the great exhibition to London 2012*. UCL Press.
- Geffroy, D., Oliver, R., Juran, L., & Skuzinski, T. (2021). Projecting the Metropolis: Paris 2024 and the (re) scaling of metropolitan governance. *Cities*, 114, Article 103189.
- Gunton, T. (2003). Megaprojects and regional development: Pathologies in project planning. *Regional Studies*, 37(5), 505–519.
- Hadjimichalis, C. (2013). From streets and squares to radical political emancipation? Resistance lessons from Athens during the crisis. *Human Geography*, 6(2), 116–136.
- Hirschman, A. O. (1967). The principle of the hiding hand. *National Affairs*, Winter 1967, 6–10.
- Hughes, N. (2018). 'Tourists go home': Anti-tourism industry protest in Barcelona. *Social Movement Studies*, 17(4), 471–477.
- Ika, A. L. (2018). Beneficial or detrimental ignorance: The straw man fallacy of Flyvbjerg's test of Hirschman's hiding hand. *World Development*, 103(2018), 369–382.
- Joly, P.-B. (2010). On the economics of techno-scientific promises. In Y. Madeline Akrich, F. M. Barthe, & P. Mustar (Eds.), *Débordements. Mélanges offerts à Michel Callon* (pp. 203–222). Paris: Presse des Mines.
- Kassens-Noor, E. (2013). Transport legacy of the Olympic games, 1992–2012. *Journal of Urban Affairs*, 35, 393–416.
- Kassens-Noor, E., Vertalka, J., & Wilson, M. (2019). Good games, bad host? Using big data to measure public attention and imagery of the Olympic games. *Cities*, 90, 229–236.
- Kennelly, J. (2015). 'You're making our city look bad': Olympic security, neoliberal urbanization, and homeless youth. *Ethnography*, 16(1), 3–24.
- Kennett, C., & Moragas, d. M. (2006). Barcelona 1992: Evaluating the Olympic legacy. In A. Tomlinson, & C. Young (Eds.), *National identity and global sports events: culture, politics, and spectacle in the Olympics and the football world cup* (pp. 177–195). Albany: State University of New York Press.
- Lagonigro, R., Martori, J. C., & Aparicio, P. (2018). Environmental noise inequity in the city of Barcelona. *Transportation Research Part D: Transport and Environment*, 63, 309–319.
- Lauermann, J. (2022). The declining appeal of megaevents in entrepreneurial cities: From Los Angeles 1984 to Los Angeles 2028. *Environment and Planning C: Politics and Space*. <https://doi.org/10.1177/23996544211066101>
- Lehtonen, M. (2014). Evaluating megaprojects: From the "iron triangle" to network mapping. *Evaluation*, 20(3), 278–295.
- Lehtonen, M. (2019). Ecological economics and opening up of megaproject appraisal: Lessons from megaproject scholarship and topics for a research programme. *Ecological Economics*, 159(May), 148–156.
- Lenskyj, H. (2000). *Inside the Olympic industry: Power, politics, and activism*. Albany, NY: State University of New York Press.
- Lenskyj, H., & Wagg, S. (2012). *The Palgrave handbook of Olympic studies*. London: Springer. <https://doi.org/10.1057/9780230367463>
- Maragall, P. (1988). *Urbanisme a Barcelona, Plans Cap el 92*. Barcelona: Ajuntament de Barcelona, Àrea de Urbanisme i Obres Públiques.
- Marshall, T. (1990). Letter from Barcelona. *Planning Practice and Research*, 5(3), 25–28.
- Marshall, T. (Ed.). (2004). *Transforming Barcelona*. Psychology Press.
- Martorell, J., Bohigas, O., Mackay, D., & Puigdomenech, A. (1992). *La Villa Olímpica: Arquitectura, Parques, Puerto Deportivo*. Barcelona: Editorial Gustavo Gili, S.A.
- Mauri, L., & Urià, L. (1998). *La Gota Malaia* (p. 62). Barcelona: Edicions 62.
- McNeill, D. (2005). *Urban change and the European left: Tales from the new Barcelona*. London: Routledge.
- Milano, C., Novelli, M., & Cheer, J. M. (2019). Overtourism and tourismophobia: A journey through four decades of tourism development, planning and local concerns. *Tourism Planning & Development*, 16(4), 353–357.
- Moix, L. (1994). *La Ciudad de los Arquitectos*. Barcelona: Crónicas Anagrama.
- Müller, M. (2015). What makes an event a mega-event? Definitions and sizes. *Leisure Studies*, 34(6), 627–642. <https://doi.org/10.1080/02614367.2014.993333>
- Müller, M., & Gafney, C. (2018). Comparing the Urban Impacts of the FIFA World Cup and Olympic Games From 2010 to 2016. *Journal of Sport and Social Issues*, 42(4), 247–269.
- Pareja-Eastaway, M., & Pique, J. M. (2011). Urban regeneration and the creative knowledge economy: The case of 22@ in Barcelona. *Journal of Urban Regeneration & Renewal*, 4(4), 319–327.
- Perdigueró, J., & Sanz, A. (2020). Cruise activity and pollution: The case of Barcelona. *Transportation Research Part D: Transport and Environment*, 78, Article 102181.
- Pitsis, A., Clegg, S., Freeder, D., Sankaran, S., & Burdon, S. (2018). Megaprojects redefined-complexity vs cost and social imperatives. *International Journal of Managing Projects in Business*, 11(1), 7–34.
- Priemus, H. (2010). Mega-projects: Dealing with pitfalls. *European Planning Studies*, 18(7), 1023–1039.
- Raventós, F. (1998). *Més d'una dècada de planificació estratègica de ciutats. 10 Anys de Planificació Estratègica a Barcelona (1988-1998)*. Barcelona: Associació Pla Estratègic Barcelona 2000 (pp. 17–28).
- Raventós, F. (2000). *La Col·laboració Publicoprivada*. Barcelona: Aula Barcelona.
- Riding, A. (1992). Barcelona, a floating game of sorts The New York Times. <https://www.nytimes.com/1992/07/29/sports/barcelona-a-floating-game-of-sorts.html>.
- Roche, M. (2002). *Megaevents and modernity: Olympics and expos in the growth of global culture*. London: Routledge.
- Roche, M. (2017). *Megaevents and social change: Spectacle, legacy and public culture*. Manchester: Manchester University Press.
- Sabaté Bel, J., & Tironi Rodó, M. (2008). Globalización y estrategias urbanísticas: un balance del desarrollo reciente de Barcelona. *Cuaderno urbano: espacio, cultura y sociedad*, 7, 233–261.
- van Marrewijk, A., Clegg, S. R., Pitsis, T. S., & Veenswijk, M. (2008). Managing public-private megaprojects: Paradoxes, complexity, and project design. *International Journal of Project Management*, 26(6), 591–600.

- Vayá, E., Garcia, J. R., Murillo, J., Román, J., & Suriñach, J. (2018). Economic impact of cruise activity: The case of Barcelona. *Journal of Travel & Tourism Marketing*, 35(4), 479–492.
- Vickerman, R. (2017). Wider impacts of megaprojects: Curse or cure? In B. Flyvbjerg (Ed.), *Oxford handbook of megaproject management*. Oxford: Oxford University Press.
- Wang, M., & Bao, H. X. (2018). Megaevent effects on the housing market: Evidence from the Beijing 2008 Olympic games. *Cities*, 72, 207–216.
- Ward, E. J. (2021). Toward a more holistic understanding of mega transport project (MTP) success. In R. Vickerman (Ed.), *International encyclopedia of transportation* (pp. 147–153). Amsterdam: Elsevier.
- Wynn, M. G. (1979). Peripheral growth of Barcelona in the Franco era. *Iberian Studies*, 8 (1), 13–28.